

EX NO:2	Chebyshev IIR Filter
DATE:	2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

```

clc;
clear;
close all;

% Filter Specifications
Wp = 0.4;    % Passband Edge Frequency (normalized)
Ws = 0.55;   % Stopband Edge Frequency (normalized)
Rp = 1;      % Passband Ripple (dB)
As = 40;     % Stopband Attenuation (dB)

% Calculate Filter Order and Cutoff Frequency
[N, Wn] = cheb1ord(Wp, Ws, Rp, As);

% Design Chebyshev Type-I Low-Pass Filter
[b, a] = cheby1(N, Rp, Wn);

% Display Results
fprintf('Filter Order = %d\n', N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n', Wn);

% Stability Check
if all(abs(roots(a)) < 1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% Frequency Response
figure;
freqz(b, a);

```

```
title('Frequency Response');
```

```
% Impulse Response  
figure;  
impz(b, a);  
title('Impulse Response');  
grid on;
```

```
% Group Delay  
figure;  
grpdelay(b, a);  
title('Group Delay');  
grid on;
```

```
% Pole-Zero Plot  
figure;  
zplane(b, a);  
title('Pole-Zero Plot');
```

Filter Order = 6
Cutoff Frequency = $0.40 * \pi$ rad/sample
System is STABLE



